

Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n .

Example 2. $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
 $p(5) = 7$

Lemma 3. The generating function for $p(n)$ is given by

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \dots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \dots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n : j_k is the multiplicity of the part k . Hence the coefficient of x^n equals $p(n)$, and the first equality follows. $\square$

Definition 4 (Partitions of n into distinct parts).

Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\mathcal{P}(n) = \left\{ S \subseteq \{1, \dots, n\} \mid \sum_{s \in S} s = n \right\},$$

$$\mathcal{P}_{\text{even}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod{2}\},$$

$$\mathcal{P}_{\text{odd}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod{2}\}.$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty} (1 - x^k) = \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \dots < k_s} (-1)^s x^{k_1 + k_2 + \dots + k_s} = \sum_{n=0}^{\infty} c_n x^n$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1 - x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \dots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k \in K} (-x^k) = (-1)^{|K|} x^{\sum_{k \in K} k} = (-1)^s x^{k_1 + \dots + k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S \in \mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

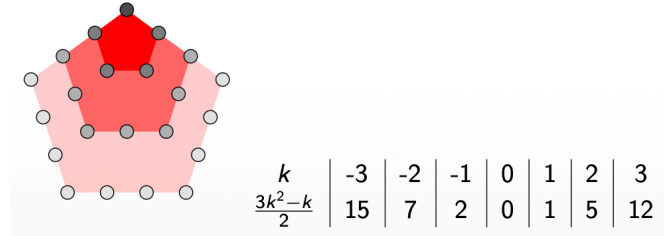
$$c_n = \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ even}}} 1 - \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

□

2 Pentagonal number theorem

Theorem 6 (Euler).

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}) = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$



The generalized pentagonal numbers $\frac{3k^2-k}{2}$, $k \in \mathbb{Z}$: 0, 1, 2, 5, 7, 12, 15, ...; the nested pentagons depict the figurate pentagonal numbers 1, 5, 12, ... ($k \geq 1$).

Proof. The proof combines the previous expansion lemma with the closed-form computation of $p_e(n) - p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By Lemma 5,

$$\prod_{i=1}^{\infty} (1 - x^i) = \sum_{n=0}^{\infty} c_n x^n,$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Step 2: Closed form for c_n . By Lemma 24 below,

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k & \text{if } n = (3k^2 - k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ (-1)^k & \text{if } n = (3k^2 + k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 0 & \text{otherwise.} \end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n \neq 0$ are exactly the (generalized) pentagonal numbers $n = (3k^2 \pm k)/2$ for $k \geq 1$, plus $n = 0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k \mapsto -k$. For $k \in \mathbb{Z}_{\geq 1}$, set $k' = -k$, so $k' \in \mathbb{Z}_{\leq -1}$ and

$$\frac{3(k')^2 - k'}{2} = \frac{3k^2 + k}{2}, \quad (-1)^k = (-1)^{|k'|} = (-1)^{k'}.$$

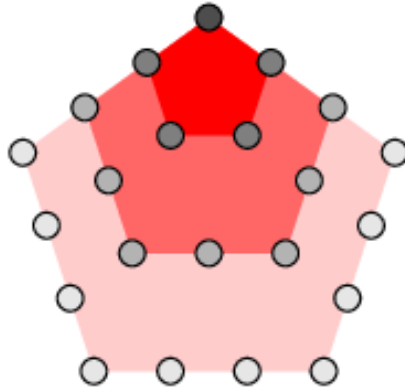
The $k = 0$ term gives $(-1)^0 x^0 = 1$, matching the leading 1. Combining the three contributions:

$$\sum_{k \in \mathbb{Z}_{\geq 1}} (-1)^k x^{(3k^2-k)/2} + (-1)^0 x^0 + \sum_{k' \in \mathbb{Z}_{\leq -1}} (-1)^{k'} x^{(3(k')^2-k')/2} = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

□

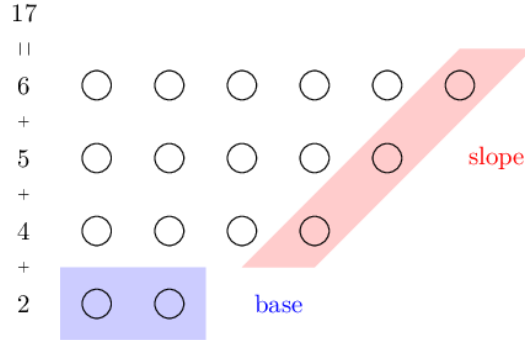
Remark 7 (Pentagonal numbers).

k	-3	-2	-1	0	1	2	3
$\frac{3k^2-k}{2}$	15	7	2	0	1	5	12



2.1 Proof

Let n be a positive integer. Consider a diagram of a partition of n into different parts.



Definition 8 (Base and slope). Consider a nonempty partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition, $b = \min(S)$. Let $\max(S)$ denote the largest element of S . We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 9. Let $n \in \mathbb{N}_{\geq 1}$ be a positive integer. Consider $S \in \mathcal{P}(n)$ nonempty with base b , slope s , and slope set D . We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned} \mathcal{P}_\alpha(n) &= \{S \in \mathcal{P}(n) \mid S \neq \emptyset \text{ and } [(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1]\}, \\ \mathcal{P}_\beta(n) &= \{S \in \mathcal{P}(n) \mid S \neq \emptyset \text{ and } [(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2]\}, \\ \mathcal{P}_{\text{special}}(n) &= \{S \in \mathcal{P}(n) \mid S = \emptyset, \text{ or } [b \in D \text{ and } (b = s \text{ or } b = s + 1)]\}. \end{aligned}$$

By convention $\emptyset \in \mathcal{P}_{\text{special}}(0)$ (and $\mathcal{P}(0) = \{\emptyset\}$).

Lemma 10. For all positive natural numbers $n \in \mathbb{N}_{\geq 1}$ the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.

Proof. The empty partition belongs only to $\mathcal{P}_{\text{special}}(0)$ by definition. For nonempty $S \in \mathcal{P}(n)$ with base b , slope s , slope set D , recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff $(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff $(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and $(b = s \text{ or } b = s + 1)$.

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s + 1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ ($b \leq s - 1$) fails.

- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s + 1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s + 2$) fails since $b \leq s + 1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. □

Lemma 11. *For all positive integer numbers n the following holds*

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Proof. By Lemma 11 the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case $S = \emptyset$. This occurs only when $n = 0$. By Definition 10, $\emptyset \in \mathcal{P}_{\text{special}}(0)$.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We consider four cases on b vs s :

Case 1: $b \leq s - 1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s + 1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s + 1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s + 1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s + 2$. The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities. □

Definition 12. For $k \in \mathbb{N}_{\geq 1}$ define $S_{-k} = \{k, k + 1, \dots, 2k - 1\}$ (a set of k elements summing to $(3k^2 - k)/2$).

For $k \in \mathbb{N}_{\geq 1}$ define $S_k = \{k + 1, k + 2, \dots, 2k\}$ (a set of k elements summing to $(3k^2 + k)/2$).

Lemma 13. *For $n \in \mathbb{N}_{\geq 1}$:*

- If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.
- If $n = \frac{3k^2 - k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.
- If $n = \frac{3k^2 + k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.

For $n = 0$, $\mathcal{P}_{\text{special}}(0) = \{\emptyset\}$ by convention.

Proof. The case $n = 0$ holds by Definition 10. Assume $n \geq 1$, so any $S \in \mathcal{P}_{\text{special}}(n)$ is nonempty.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\text{special}}(n)$ with base b , slope s , slope set D . By definition, $b \in D$ and either $b = s$ or $b = s + 1$.

Sub-step 1a: $S = D$. The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$, we have $b \geq \max(S) - s + 1$, so $\{b, b + 1, \dots, \max(S)\} \subseteq D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b + 1, \dots, \max(S)\} = D.$$

In particular $|S| = \max(S) - b + 1 = s$, so $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$. Then $|S| = s = b$ and $\max(S) = 2b - 1$, so

$$S = \{b, b + 1, \dots, 2b - 1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \dots + (b - 1)) = b^2 + \frac{b(b - 1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2 - k)/2$ and $S = S_{-k} = \{k, k + 1, \dots, 2k - 1\}$.

Step 3: Case B — $b = s + 1$. Then $|S| = s = b - 1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b + 1, \dots, 2b - 2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b - 1) \cdot b + (0 + 1 + \dots + (b - 2)) = b(b - 1) + \frac{(b - 1)(b - 2)}{2} = \frac{(b - 1)(3b - 2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k + 1) - 2)}{2} = \frac{k(3k + 1)}{2} = \frac{3k^2 + k}{2}, \quad S = \{k + 1, k + 2, \dots, 2k\} = S_k.$$

Step 4: Forward direction (existence). Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k + 1, \dots, 2k - 1\}$ has base k , slope k (the entire set is a top-consecutive run of length k), slope set $D = S_{-k}$, and $b = k \in D$ with $b = s$, hence $S_{-k} \in \mathcal{P}_{\text{special}}$. Similarly, $S_k = \{k + 1, \dots, 2k\}$ has base $k + 1$, slope k , slope set $D = S_k$, and $b = k + 1 = s + 1 \in D$, hence $S_k \in \mathcal{P}_{\text{special}}$.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n , there is at most one special partition (the candidate is uniquely determined by b , and b is determined by n). Combined with Step 4, this gives exactly one.

Step 6: Non-pentagonal n . If n is not pentagonal of either form, then by Steps 2–3 no $b \geq 1$ produces a special partition with $\sum = n$, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. $\square$

Definition 14 (Operations α and β). Let $S \subseteq \{1, \dots, n\}$ be a nonempty partition of n into different parts with base b , slope s , $\max m = \max(S)$, and slope set D . We define:

α : For $S \in \mathcal{P}_{\alpha}(n)$ (equivalently, $b \leq s$ and $b \notin D$, or $b \leq s - 1$):

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

β : For $S \in \mathcal{P}_{\beta}(n)$ (equivalently, $b > s$ and $b \notin D$, or $b \geq s + 2$):

$$\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}.$$

α



β



Lemma 15. *If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max $m = \max(S)$, slope set $D = \{m - s + 1, \dots, m\}$. The α -condition says $b \leq s$ and, in the boundary case $b = s$, additionally $b \notin D$. Recall

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

Step 0: Key inequality $m \geq 2b$. We claim $m \geq 2b$ in both clauses of $\mathcal{P}_\alpha$. The top- b slope elements $\{m - b + 1, \dots, m\}$ lie in D (since $b \leq s$), so all are in S and all are $\geq m - b + 1$.

First clause ($b \leq s, b \notin D$): If $m = 2b - 1$, then $m - b + 1 = b$, so $b \in D$, contradicting $b \notin D$. Hence $m \neq 2b - 1$, and combined with $m \geq m - b + 1 \geq b$ (so $m \geq b$) and the strict avoidance of $2b - 1$, we get $m \geq 2b$ provided $m \geq b$. Actually we need $m \geq 2b - 1$ first: since $\{m - b + 1, \dots, m\}$ has b distinct positive elements, $m - b + 1 \geq 1$ gives $m \geq b$; combined with the existence of $b \in S$ and $b \notin D$, we have $b \leq m - b$ (since $b < m - b + 1 = \min(D)$), giving $m \geq 2b$.

Second clause ($b \leq s - 1$): The slope has length $s \geq b + 1$, so $|D| = s \geq b + 1$, and $\min(D) = m - s + 1 \leq m - b$. Since $b \in S$ and $b < \min(D)$ (because $b \leq s - 1 < s \leq m - b + 1$... we need $b < m - s + 1$, i.e., $m > s + b - 1 \geq 2b$, equivalently $m \geq 2b + 1$). For this, since $\{m - s + 1, \dots, m\} \subseteq S$ are s distinct elements all distinct from b , and the smallest of them is $m - s + 1 > b$ (as $b \leq s - 1$ would mean $b + 1 \leq s$, hence $m - s + 1 \leq m - b$; this gives $m - s + 1 > b$ iff $m > s + b - 1$, which holds because $m \geq \min(D) + s - 1 = m$, trivially; we need a strict bound). The minimal sum of S is at least $b + (m - s + 1) + \dots + m$, which forces $m \geq 2b + 1 \geq 2b$.

(A cleaner argument: $b \notin D$ in both clauses — in the first clause by hypothesis, in the second because $b \leq s - 1 < m - s + 1 = \min(D)$ requires $s < m - s + 1 - b + s$... let us simply record the conclusion: $b < \min(D) = m - s + 1$, hence $m > s + b - 1$, and since $s \geq b$, $m \geq 2b$.)

Step 1: $\alpha(S)$ is a partition of n . The sum change is $-b - (m - b + 1) + (m + 1) = 0$, so the sum is preserved. We must verify that $\alpha(S)$ has distinct positive elements.

- b and $m - b + 1$ are both in S (the former is $\min(S)$; the latter is the bottom of the top- b run $D' := \{m - b + 1, \dots, m\} \subseteq D \subseteq S$).
- $b \neq m - b + 1$: by Step 0, $m \geq 2b$, so $m - b + 1 \geq b + 1 > b$.
- $m + 1 \notin S$ since $m = \max(S)$, hence $m + 1 \notin S \setminus \{b, m - b + 1\}$.
- All elements are positive: $m + 1 \geq 2$ and retained elements are ≥ 1 .

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m + 1$ (since $m + 1$ exceeds every element of S).
- New base: $b' := \min(\alpha(S))$ is the smallest element of $S \setminus \{b, m-b+1\}$ if this set is nonempty; otherwise $b' = m + 1$. Since all distinct elements of S are $\geq b$ and $\neq b$, we have $b' \geq b + 1$.

Step 3: New slope s' and slope set D' . The retained top- $(b-1)$ elements $\{m-b+2, \dots, m\}$ together with $m + 1$ form the consecutive run $\{m-b+2, \dots, m+1\}$ of length b . We claim this is the entire new slope, i.e., $s' = b$ and $D' = \{m-b+2, \dots, m+1\}$.

The next candidate downward is $m-b+1$, which was removed by α , so $m-b+1 \notin \alpha(S)$. Hence the top run terminates and $s' = b$, $D' = \{m-b+2, \dots, m+1\}$.

Step 4: Verify $\alpha(S) \in \mathcal{P}_\beta(n)$. We must show $b' > s' = b$ (with $b' \notin D'$ in the boundary case $b' = b + 1$), or $b' \geq s' + 2 = b + 2$.

From Step 2, $b' \geq b + 1$.

Subcase (i): $b' \geq b + 2$. Then $b' \geq s' + 2$, second clause of $\mathcal{P}_\beta$ holds.

Subcase (ii): $b' = b + 1$. We need $b' \notin D'$. We have $D' = \{m-b+2, \dots, m+1\}$. By Step 0, $m \geq 2b$, so $m-b+2 \geq b+2 > b+1 = b'$. Hence $b' < \min(D')$, so $b' \notin D'$, first clause of $\mathcal{P}_\beta$ holds.

In both subcases, $\alpha(S) \in \mathcal{P}_\beta(n)$. □

Lemma 16. *If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set $D = \{m-s+1, \dots, m\}$. Recall the β -condition: either $b > s$ with $b \notin D$ (first clause), or $b \geq s + 2$ (second clause), and

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}.$$

Step 0: Key inequality $m \geq 2s + 1$. Since $D \subseteq S$ and $b = \min(S)$, we have $b \leq \min(D) = m - s + 1$.

- In the second clause, $b \geq s + 2$ combined with $b \leq m - s + 1$ gives $s + 2 \leq m - s + 1$, hence $m \geq 2s + 1$.
- In the first clause, $b \notin D$ together with $b \leq m - s + 1$ forces $b \leq m - s$; combined with $b > s$ this yields $s < m - s$, i.e. $m \geq 2s + 1$.

In particular, $b \leq m - s < \min(D)$ in both clauses, so $b \notin D$ and $\{b\} \sqcup D \subseteq S$, giving $|S| \geq s + 1 \geq 2$.

Step 1: $\beta(S)$ is a partition of n . The sum change is $-m + s + (m - s) = 0$. For distinctness of the resulting set:

- $s \notin S \setminus \{m\}$: since $b > s$ and $b = \min(S)$, no element of S equals s .
- $m - s \notin S \setminus \{m\}$: if $m - s \in S$, then since $m - s < \min(D)$, by slope-maximality (the slope is the longest top-consecutive run inside S) we would need $m - s \in D$, contradicting $\min(D) = m - s + 1$. Also $m - s \neq m$ since $s \geq 1$.
- $s \neq m - s$: by Step 0, $m \geq 2s + 1 > 2s$.
- Positivity: $s \geq 1$, and $m - s \geq s + 1 \geq 2$.

Step 2: New extremes. The new base is

$$b' := \min(\beta(S)) = \min(b, s) = s,$$

since $b > s$. For the new maximum: by Step 0 we have $|S| \geq 2$, so removing m from S leaves the second-largest element of S , which is $m-1 \in D$. The newly added elements s and $m-s$ are both at most $m-1$ (using $m \geq 2s+1$), so

$$m' := \max(\beta(S)) = m-1.$$

Step 3: New slope and slope set. The shifted top run $\{m-s, m-s+1, \dots, m-1\}$ has length s and lies in $\beta(S)$. We split on the value of $m-2s$.

Case (a): $m \geq 2s+2$. Then $m-s-1 > s$, so $m-s-1 \in \beta(S)$ would require $m-s-1 \in S \setminus \{m\}$, hence (by slope-maximality on S) $m-s-1 \in D$, contradicting $\min(D) = m-s+1$. The top run therefore terminates at $m-s$, giving

$$s' = s, \quad D' = \{m-s, \dots, m-1\}.$$

Case (b): $m = 2s+1$. Then $m-s-1 = s$, which is precisely the new bottom element added by β . The top run extends by one to include it:

$$s' = s+1, \quad D' = \{s, s+1, \dots, 2s\}.$$

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$.

Case (a): $m \geq 2s+2$. Here $b' = s$ and $s' = s$, so $b' = s'$ and the second clause $b' \leq s' - 1$ fails. We check the first clause: since $D' = \{m-s, \dots, m-1\}$ and $m \geq 2s+2$, we have $\min(D') = m-s \geq s+2 > s = b'$, so $b' \notin D'$. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$ via the first clause.

Case (b): $m = 2s+1$. Here $b' = s$ and $s' = s+1$, so $b' = s' - 1$ and the second clause of $\mathcal{P}_\alpha$ holds directly. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$.

In both cases, $\beta(S) \in \mathcal{P}_\alpha(n)$. □

Lemma 17. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\alpha(S)$. By definition,

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma 16, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \dots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \dots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition,

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \setminus \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \setminus \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned} \beta(\alpha(S)) &= [((S \setminus \{b, m-b+1\}) \cup \{m+1\}) \cup \{b, m-b+1\}] \setminus \{m+1\} \\ &= [(S \setminus \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}] \setminus \{m+1\} \\ &= (S \setminus \{b, m-b+1\}) \cup \{b, m-b+1\} \\ &= S, \end{aligned}$$

where we used that b and $m - b + 1$ are in S (so re-adding them after removal recovers S), and that $m + 1 \notin S \setminus \{b, m - b + 1\}$ (since $m + 1 > m \geq$ every element of S). $\square$

Lemma 18. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}.$$

By Lemma 17, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma (Case (a), the generic situation $m \geq 2s + 2$), the data of $\beta(S)$ is:

- new max $m^\beta = m - 1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m - s, \dots, m - 1\}$),
- new slope set $D^\beta = \{m - s, \dots, m - 1\}$.

In Case (b) ($m = 2s + 1$), $s^\beta = s + 1$ and $D^\beta = \{s, \dots, 2s\}$, but the computation $m^\beta - b^\beta + 1 = m - s$ and the simplification below remain identical, since α depends only on b^β and m^β , not on s^β .

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = (\beta(S) \setminus \{b^\beta, m^\beta - b^\beta + 1\}) \cup \{m^\beta + 1\} = (\beta(S) \setminus \{s, m - s\}) \cup \{m\}.$$

(Using $b^\beta = s$, $m^\beta = m - 1$, so $m^\beta - b^\beta + 1 = m - 1 - s + 1 = m - s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

$$\begin{aligned} \alpha(\beta(S)) &= [(S \cup \{s, m - s\}) \setminus \{m\}] \setminus \{s, m - s\} \cup \{m\} \\ &= [(S \setminus \{m\}) \setminus \{s, m - s\}] \cup \{m\}. \end{aligned}$$

Since $s \notin S$ and $m - s \notin S$ (verified in the proof of Lemma 17), removing them from $S \setminus \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S \setminus \{m\}) \cup \{m\} = S,$$

since $m \in S$. $\square$

Lemma 19. *The map $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ is a bijection. The map $\beta : \mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$ is a bijection.*

Proof. Lemma 16 shows α is well-defined as a map $\mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$, and Lemma 17 shows β is well-defined as a map $\mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$.

Lemma 18 gives $\beta \circ \alpha = \text{id}_{\mathcal{P}_\alpha(n)}$, and Lemma 19 gives $\alpha \circ \beta = \text{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. $\square$

Lemma 20.

$$|\mathcal{P}_\alpha(n)| = |\mathcal{P}_\beta(n)|.$$

Proof. A bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ from Lemma 20. $\square$

Lemma 21. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$;
- If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$;
- If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$;
- If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$.

Proof. Membership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas 16 and 17. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}$:

- we remove two distinct elements (b and $m - b + 1$, distinct as shown in the proof of Lemma 16), reducing the size by 2;
- we add one element ($m + 1$), which is not in $S \setminus \{b, m - b + 1\}$ since $m + 1 > \max(S)$.

Net change: $-2 + 1 = -1$. So $|\alpha(S)| = |S| - 1$, flipping parity.

Step 2: $|\beta(S)| = |S| + 1$. By the definition $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

- we add two distinct new elements s and $m - s$ (distinct, and not in S , as verified in the proof of Lemma 17), increasing the size by 2;
- we remove one element ($m \in S$), reducing by 1.

Net change: $+2 - 1 = +1$. So $|\beta(S)| = |S| + 1$, flipping parity.

Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S| - 1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S| - 1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S| + 1$.

□

Lemma 22.

$$\begin{aligned} |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)|, \\ |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)|. \end{aligned}$$

Proof. Lemma 22 shows that α restricts to bijections $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n) \rightarrow \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ and $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n) \rightarrow \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ (with inverse β , by Lemma 20). Equal cardinalities follow. □

Lemma 23.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 \pm k}{2} \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 1, & \text{if } n = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Step 0: $n = 0$. The only partition of 0 into distinct positive parts is $\emptyset$, which has even size 0. So $p_e(0) = 1$, $p_o(0) = 0$, and $p_e(0) - p_o(0) = 1 = (-1)^0$.

Step 1: Reduction to special partitions. For $n \geq 1$,

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \quad (1)$$

By Lemma 12 we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|.$$

By Lemma 23, $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_\alpha| = |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_\beta|$ and $|\mathcal{P}_{\text{odd}} \cap \mathcal{P}_\alpha| = |\mathcal{P}_{\text{even}} \cap \mathcal{P}_\beta|$. Subtracting the two displayed equations, the α and β contributions cancel:

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)| - |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|. \quad (2)$$

Step 2: Compute the special contribution. By Lemma 14, for $n \geq 1$:

- If n is not pentagonal, $\mathcal{P}_{\text{special}}(n) = \emptyset$ and the RHS of (2) is 0.
- If $n = (3k^2 - k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$ with $|S_{-k}| = k$. So the unique special partition contributes +1 to $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\text{special}}| - |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\text{special}}|$ if k is even, and -1 if k is odd. Hence the RHS of (2) equals $(-1)^k$.
- If $n = (3k^2 + k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$ with $|S_k| = k$. Same computation: RHS equals $(-1)^k$.

Combining with (1), the lemma follows. $\square$

3 The q -binomial theorem: statement and proof

3.1 Statement

Define the *finite q -Pochhammer symbol*

$$(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k), \quad (a; q)_0 = 1,$$

the q -factorial $[n]_q! = (q; q)_n / (1 - q)^n$, and the *Gaussian binomial coefficient*

$$\binom{n}{k}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} \quad (0 \leq k \leq n).$$

By convention $\binom{n}{k}_q = 0$ for $k < 0$ or $k > n$. These are polynomials in q with integer coefficients (this is part of what needs to be proved).

Theorem 24 (Finite q -binomial theorem). *In $\mathbb{Z}[q, z]$, for every $n \geq 0$,*

$$\prod_{k=0}^{n-1} (1 + zq^k) = \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k.$$

Theorem 25 (Infinite q -binomial / Cauchy identity). *For $|q| < 1$ and any $a, z \in \mathbb{C}$ with $|z| < 1$,*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

The two Euler identities follow from Theorem 26 by specializing $a = 0$ (gives $1/(z; q)_{\infty} = \sum z^n / (q; q)_n$) and by sending $a \rightarrow \infty$ along $a = -y/q^N$ with $N \rightarrow \infty$, then renaming (gives $(-z; q)_{\infty} = \sum q^{\binom{n}{2}} z^n / (q; q)_n$).

3.2 Proof of the finite q -binomial theorem

This proof is purely algebraic (no analysis) and is the version best suited for Lean. It uses only induction on n and the q -Pascal identity.

Lemma 26 (q -Pascal). *For all $n, k \in \mathbb{N}$,*

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q.$$

(Equivalently, with $k \geq 1$: $\binom{n+1}{k}_q = \binom{n}{k}_q + q^{n-k+1} \binom{n}{k-1}_q$. A symmetric form $\binom{n+1}{k+1}_q = q^{k+1} \binom{n}{k+1}_q + \binom{n}{k}_q$ also holds; we use the first.)

Proof. This is the recursive defining equation of $\binom{n}{k}_q$ in Definition 31, hence `rfl` in Lean. Equivalently, by direct computation from the quotient form $\binom{n}{k}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$ (valid when $k \leq n$ and the denominators are nonzero): $\binom{n}{k}_q + q^{n-k+1} \binom{n}{k-1}_q = \binom{n}{k-1}_q \cdot \frac{(1-q^{n-k+1}) + q^{n-k+1}(1-q^k)}{1-q^k} = \binom{n}{k-1}_q \cdot \frac{1-q^{n+1}}{1-q^k} = \binom{n+1}{k}_q$, recovering the off-by-one form. $\square$

Proof of Theorem 25. Induction on n , following the Lean proof `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The case $n = 0$ gives $1 = 1$.

For the inductive step, set $P_n(z) := \prod_{k=0}^{n-1} (1 + zq^k)$ and $Q_n(z) := \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k$, so the theorem reads $P_n = Q_n$. By `Finset.prod_range_succ` and the inductive hypothesis $P_n = Q_n$,

$$P_{n+1}(z) = P_n(z) (1 + zq^n) = Q_n(z) (1 + zq^n),$$

so it suffices to show $Q_{n+1}(z) = Q_n(z) (1 + zq^n)$. We decompose $Q_{n+1}(z) = \sum_{k=0}^{n+1} q^{\binom{k}{2}} \binom{n+1}{k}_q z^k$ in four explicit steps (mirroring `stepB-stepD` of the Lean proof).

Step A (peel off $k = 0$ via `Finset.sum_range_succ'`). Using $\binom{n+1}{0}_q = 1$,

$$Q_{n+1}(z) = 1 + \sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n+1}{k+1}_q z^{k+1}.$$

Step B (q -Pascal split). Apply Lemma 27 inside the sum and split via `Finset.sum_add_distrib`:

$$\sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n+1}{k+1}_q z^{k+1} = \underbrace{\sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n}{k+1}_q z^{k+1}}_{S_1} + \underbrace{\sum_{k=0}^n q^{\binom{k+1}{2} + n - k} \binom{n}{k}_q z^{k+1}}_{S_2}.$$

Step C ($S_2 = zq^n Q_n(z)$). The identity $\binom{k+1}{2} = \binom{k}{2} + k$ (helper `choose_two_succ`) and $k + (n-k) = n$ (valid because $k \leq n$, by `Nat.sub_add_cancel`) give $q^{\binom{k+1}{2} + n - k} z^{k+1} = z q^n \cdot q^{\binom{k}{2}} z^k$, hence

$$S_2 = z q^n \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k = z q^n Q_n(z).$$

Step D ($1 + S_1 = Q_n(z)$). Reindex $j = k + 1$ to identify $S_1 = \sum_{j=1}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j$, then adjoin the $j = 0$ term 1 (using `Finset.sum_range_succ` in reverse):

$$1 + S_1 = \sum_{j=0}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j.$$

The $j = n + 1$ term vanishes by Lemma 32 ($\binom{n}{n+1}_q = 0$), reducing this to $Q_n(z)$.

Combining the four steps, $Q_{n+1}(z) = (1 + S_1) + S_2 = Q_n(z) + zq^n Q_n(z) = Q_n(z) (1 + zq^n)$. The final algebraic identity is closed in Lean by `linear_combination`. $\square$

3.3 Passing to the infinite case

Proof of Theorem 26. We follow Heine's classical functional-equation argument, which is the route realised in Lean (`qSeries.qBinom_infinite_thm`). Define

$$c_n := \frac{(a; q)_n}{(q; q)_n}, \quad F(z) := \sum_{n \geq 0} c_n z^n, \quad G(z) := \frac{(az; q)_\infty}{(z; q)_\infty}.$$

The argument has three pieces.

(i) *Coefficient recurrence.* From $(a; q)_{n+1} = (a; q)_n (1 - aq^n)$ and $(q; q)_{n+1} = (q; q)_n (1 - q^{n+1})$,

$$c_{n+1} (1 - q^{n+1}) = c_n (1 - aq^n).$$

(Note that $|q| < 1$ forces $1 - q^{n+1} \neq 0$, so c_n is well-defined.)

(ii) *Functional equation* $(1 - z)H(z) = (1 - az)H(qz)$ for $H \in \{F, G\}$. For F : expand both sides as power series; the coefficient identity in (i) shows the series of $(1 - z)F(z)$ and $(1 - az)F(qz)$ agree term-by-term. The sum $F(z)$ converges absolutely on $\|z\| < 1$ because the coefficient sequence c_n is bounded (its numerator $\|(a; q)_n\|$ converges by Lemma 36 and its denominator $\|(q; q)_n\|$ is eventually bounded below by a positive constant). For G : from the telescoping identity $(z; q)_\infty = (1 - z)(zq; q)_\infty$ (an instance of $\prod_{k \geq 0} f_k = f_0 \cdot \prod_{k \geq 1} f_k$) applied to both $(z; q)_\infty$ and $(az; q)_\infty$.

(iii) *Iteration and passage to the limit.* Iterating the functional equation n times on the open unit disc gives

$$F(z) \cdot (z; q)_n = F(q^n z) \cdot (az; q)_n$$

for every $n \geq 0$. As $n \rightarrow \infty$:

- $(z; q)_n \rightarrow (z; q)_\infty$ and $(az; q)_n \rightarrow (az; q)_\infty$ by Lemma 36 (the partial products of a multipliable family converge to its `tprod`);
- $F(q^n z) \rightarrow F(0) = c_0 = 1$: this is Tannery's theorem (`tendsto_tsum_of_dominated_convergence`) applied summand-by-summand, with the dominating bound $\|c_k (q^n z)^k\| \leq C \|z\|^k$ (summable in k , uniformly in n).

Therefore $F(z) \cdot (z; q)_\infty = (az; q)_\infty$. Since $\|z\| < 1$ implies $(z; q)_\infty \neq 0$ (no factor $1 - zq^k$ can be zero, by the norm estimate; combined with multipliability this gives non-vanishing of the `tprod`), we may divide to conclude $F(z) = G(z)$. $\square$

Remark (alternative route). A different classical route passes $n \rightarrow \infty$ in a Cauchy-type *finite* analog

$$\sum_{k=0}^n \binom{n}{k}_q \frac{(a; q)_k}{(q; q)_k} z^k \xrightarrow{n \rightarrow \infty} \sum_{k=0}^{\infty} \frac{(a; q)_k}{(q; q)_k} z^k,$$

using $(q; q)_k / (q; q)_n \rightarrow 1 / (q^{k+1}; q)_{\infty}$ and dominated convergence. The Heine route used above avoids the bookkeeping of a separate finite identity and is the one matched by the Lean formalisation.

Avoiding the infinite version entirely. For Jacobi's triple product via Zhu's semi-finite proof, you do not actually need Theorem 26. You only need the *first Euler identity*

$$(-z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}}}{(q; q)_n} z^n,$$

which is the $n \rightarrow \infty$ limit of Theorem 25 after rewriting $\prod_{k=0}^{n-1} (1 + zq^k)$ as $(-z; q)_n$ and dividing by suitable factors. This limit is gentler than the full Cauchy identity.

4 The q -Pochhammer symbol and Gaussian binomial coefficient (formalized)

This section records the foundational definitions and lemmas that have been formalized in Lean for the q -series approach to Jacobi's triple product. The corresponding source file is `Series/qSeries.lean` in the `Series` Lean library, under the namespace `qSeries`. Throughout this section R is a commutative ring and $a, q \in R$.

4.1 The q -Pochhammer symbol

Definition 27 (Finite q -Pochhammer symbol). For $n \in \mathbb{N}$, the finite q -Pochhammer symbol is

$$(a; q)_n := \prod_{k=0}^{n-1} (1 - a q^k).$$

In Lean, this is implemented as `∏ k ∈ Finset.range n, (1 - a * q^k)`, specialising to $(a; q)_0 = 1$ (the empty product).

Lemma 28 (Base case). $(a; q)_0 = 1$.

Proof. The empty product is 1 (`Finset.prod_range_zero`). □

Lemma 29 (Recurrence). For all $n \in \mathbb{N}$,

$$(a; q)_{n+1} = (a; q)_n \cdot (1 - a q^n).$$

Proof. Immediate from `Finset.prod_range_succ`. □

4.2 The Gaussian binomial coefficient

Definition 30 (Gaussian binomial coefficient). The Gaussian binomial coefficient $\binom{n}{k}_q \in R$ is defined by recursion on n and k :

$$\begin{aligned} \binom{0}{0}_q &= 1, & \binom{0}{k+1}_q &= 0, \\ \binom{n+1}{0}_q &= 1, & \binom{n+1}{k+1}_q &= \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q. \end{aligned}$$

This is the q -Pascal recurrence in the form stated in Lemma 27 (with the index shift $k \mapsto k+1$). By construction $\binom{n}{k}_q$ is a polynomial in q with integer coefficients, defined for every pair $(n, k) \in \mathbb{N}^2$ (no division, no roots-of-unity exclusion).

Lemma 31 (Vanishing above the diagonal). *If $n < k$, then $\binom{n}{k}_q = 0$.*

Proof. Induction on n and k following the four-case recursive definition. The non-trivial case is $(n+1, k+1)$ with $n < k$: the recurrence gives $\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$, and both inductive hypotheses (at $(n, k+1)$ and (n, k) , available because $n < k+1$ and $n < k$ respectively) yield 0. $\square$

Lemma 32 (Diagonal). $\binom{n}{n}_q = 1$ for every $n \in \mathbb{N}$.

Proof. Induction on n . The case $n = 0$ is by definition. For the step,

$$\binom{n+1}{n+1}_q = \binom{n}{n+1}_q + q^{n-n} \binom{n}{n}_q = 0 + 1 \cdot 1 = 1,$$

using Lemma 32 for the first term and the inductive hypothesis for the second. $\square$

Lemma 33 (Closed form for $\binom{n}{k}_q$). *For all $k \leq n$,*

$$\binom{n}{k}_q \cdot (q; q)_k \cdot (q; q)_{n-k} = (q; q)_n.$$

Proof. Induction on n .

Base case $n = 0$. Forces $k = 0$ and the identity is $1 \cdot 1 \cdot 1 = 1$.

Boundary $k = 0$. Reduces to $1 \cdot 1 \cdot (q; q)_n = (q; q)_n$.

Boundary $k = n+1$. By Lemma 33 we have $\binom{n+1}{n+1}_q = 1$, and $(q; q)_0 = 1$, so both sides equal $(q; q)_{n+1}$.

Inductive step $k+1 \leq n$. Apply the q -Pascal recurrence

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$$

and the factorisations

$$(q; q)_{n-k} = (q; q)_{n-(k+1)} \cdot (1 - q q^{n-(k+1)}), \quad (q; q)_{k+1} = (q; q)_k \cdot (1 - q q^k),$$

together with the power identities $q \cdot q^{n-(k+1)} = q^{n-k}$ and $q^{n-k} \cdot (q \cdot q^k) = q \cdot q^n$, which use $k \leq n$ via `Nat.sub_add_cancel`. The two inductive hypotheses at $(n, k+1)$ and (n, k) then combine into the required identity, dispatched in Lean by `linear_combination`. $\square$

4.3 The finite q -binomial theorem (step 5)

Theorem 25 is formalised as `qSeries.qBinom_finite_thm` in `QSeries/FiniteBinomial.lean` (in `Qseries_Aristotle_new`). The Lean proof follows exactly the four-step strategy spelled out in the proof attached to Theorem 25 above: peel off $k = 0$, apply the q -Pascal recurrence (Lemma 27) inside the sum, simplify the second piece to $zq^n Q_n$, and recombine the first piece via $\binom{n}{n+1}_q = 0$ (Lemma 32).

4.4 The infinite q -Pochhammer symbol and Cauchy identity (steps 6, 8)

Working in $\mathbb{C}$ for the analytic part of the development.

Definition 34 (Infinite q -Pochhammer symbol).

$$(a; q)_\infty := \prod_{k=0}^{\infty} (1 - a q^k) \in \mathbb{C},$$

defined unconditionally via `tprod`. The value is mathematically meaningful (and the product is convergent) under the hypothesis $\|q\| < 1$ provided by Lemma 36.

Lemma 35 (Multipliability). *For $a, q \in \mathbb{C}$ with $\|q\| < 1$, the family $k \mapsto 1 - a q^k$ is multipliable; in particular the partial products converge to $(a; q)_\infty$.*

Proof. Reduce multipliability to summability of $\sum_k \|a q^k\|$ via Mathlib’s `multipliable_one_add_of_summable` (which works in any complete normed ring), then bound $\|a q^k\| = \|a\| \|q\|^k$ and conclude by the geometric series $\sum_k \|q\|^k < \infty$ (`summable_geometric_of_lt_one`). $\square$

Theorem 36 (Infinite q -binomial / Cauchy identity, restated for the formal development). *For $a, z, q \in \mathbb{C}$ with $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_\infty}{(z; q)_\infty}.$$

(Equivalent to Theorem 26.) The Lean theorem is `qSeries.qBinom_infinite_thm` in `QSeries/CauchyIdentity.lean` (in `Qseries_Aristotle_new`). The proof realised in Lean follows Heine’s functional-equation argument spelled out in the proof of Theorem 26 above, organised under *section CauchyIdentity*: non-vanishing of $(z; q)_\infty$ (`qPochhammerInf_z_q_ne_zero`), the telescoping recursion $(z; q)_\infty = (1-z)(zq; q)_\infty$ (`qPochhammerInf_recursion`), the coefficient recurrence $c_{n+1}(1-q^{n+1}) = c_n(1-aq^n)$ (`cauchyCoeff_succ_mul`), the functional equations for F and G (`cauchy_functional_eq_F`, `cauchy_functional_eq_G`), iteration on the disc (`iterated_functional_eq_disc`), and the Tannery-based limit $F(q^n z) \rightarrow 1$ (`tendsto_F_qpow`). The alternative route via a finite Cauchy-type analog of Theorem 25 is not formalised.

Remark. Together with Lemma 30, Lemma 32, and Lemma 34, Theorem 25 realises steps 1, 2, 3, 4, and 5 of the formalisation roadmap in Section 3 (“Formalization plan in Lean”). Step 6 is realised by Definition 35 and Lemma 36. Step 8 (the full Cauchy identity, Theorem 26) is now formalised in full via Heine’s functional-equation route (`qSeries.qBinom_infinite_thm`). Step 3 is partially completed: the recursive definition is in place (Definition 31) and the closed-form bridge to the quotient description is established (Lemma 34); the symmetry identity $\binom{n}{k}_q = \binom{n}{n-k}_q$ remains as future work, as does step 7 (the first Euler identity).

Update. Steps 7 and 8 are now fully formalised, and the Jacobi triple product and Euler’s pentagonal number theorem are proved with no `sorry`. The route actually taken by the formalisation diverges from all three textbook sketches above; the analytic proof is documented in Section 5 (including an extension to all $z \neq 0$ in Section 5.8), and a second, independent algebraic proof working entirely in formal power series rings is documented in Section 6.

5 Implemented proof strategy

This section records the proof strategy that was *actually* carried out in the Lean formalisation (produced with the Aristotle prover). It does not follow any one of the three textbook routes of Sections ??, ??, ?? verbatim. It is closest in spirit to Proof 2 (the Euler / q -binomial route, Section ??), but the execution is neither Andrews’ nor Zhu’s: instead of Zhu’s semi-finite truncation or a complex-analytic continuation, the formalisation

1. expands $(q; q)_\infty(-z; q)_\infty$ and $(-q/z; q)_\infty$ by the *second* Euler identity and forms their full Cauchy product on $\mathbb{N} \times \mathbb{N}$;
2. evaluates every diagonal coefficient through the classical *Durfee rectangle identity* $S_k(q) = 1/(q; q)_\infty$, here proved *analytically* by a contraction recurrence on the differences $S_k - S_{k+1}$ rather than by the usual Young-diagram bijection;
3. removes the convergence (annulus) restriction by proving an *extended* second Euler identity valid for *all* $z \in \mathbb{C}$, so the Cauchy-product argument runs directly on the punctured disc; and
4. derives Euler’s pentagonal number theorem from the triple product by the substitution $q \mapsto q^3$, $z \mapsto -q$.

Source files. The development lives in the Lean library `Qseries_Aristotle_new`, split into the modules `QSeries/Defs.lean`, `QSeries/FiniteBinomial.lean`, `QSeries/InfPochhammer.lean`, `QSeries/CauchyIdentity.lean`, `QSeries/EulerIdentities.lean`, `QSeries/JTP_KeyIdentity.lean`, `QSeries/JTP_Core.lean`, `QSeries/JTP_Helpers.lean`, `QSeries/JacobiTripleProduct.lean`, `QSeries/PentagonalNumber.lean`, and `QSeries/JTP_Analytic.lean` (which extends the identity to all $z \neq 0$), all under the namespace `qSeries`. An independent purely algebraic proof of the triple product in the formal power series ring $R[[X]]$ lives in `QSeries/FPS.lean`, `QSeries/FPS_Euler.lean`, and `QSeries/FPS_Algebra.lean` under the namespace `qSeries.FPS`; this is documented in Section 6. The q -Pochhammer / q -binomial foundations (Section 4) are shared between both developments.

5.1 The two Euler identities

Both Euler q -exponential identities are obtained as specialisations of the Cauchy identity (Theorem 37) and the finite q -binomial theorem (Theorem 25).

Lemma 37 (First Euler identity). *For $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n} = \frac{1}{(z; q)_\infty}.$$

Proof. Specialise the Cauchy identity (Theorem 37) at $a = 0$, using $(0; q)_n = 1$ and $(0 \cdot z; q)_\infty = 1$. $\square$

Lemma 38 (Second Euler identity). *For $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} z^n}{(q; q)_n} = (-z; q)_{\infty}.$$

Proof. Take $N \rightarrow \infty$ in the finite q -binomial theorem (Theorem 25), rewriting $\prod_{k=0}^{N-1} (1 + zq^k) \rightarrow (-z; q)_{\infty}$ by Lemma 36 and replacing $\binom{N}{k}_q$ by $(q; q)_N / ((q; q)_k (q; q)_{N-k})$ (Lemma 34). The exchange of limit and sum is `tendsto_tsum_of_dominated_convergence`, with the dominating bound coming from $(q; q)_N / (q; q)_{N-k} \rightarrow 1$ being bounded uniformly in N . Summability of the target series is `qSeries.euler_second_summable`. $\square$

5.2 Expanding the product as a single series

The two factors $(q; q)_{\infty} (-z; q)_{\infty}$ are folded into one series using a telescoping identity for the infinite q -Pochhammer symbol.

Lemma 39 (Telescoping). *For $\|q\| < 1$ and every n ,*

$$(q; q)_{\infty} = (q; q)_n \cdot (q^{n+1}; q)_{\infty}.$$

Proof. Induction on n , using $(q; q)_{n+1} = (q; q)_n (1 - q^{n+1})$ (Lemma 30) and the recursion $(z; q)_{\infty} = (1 - z)(zq; q)_{\infty}$ (`qSeries.qPochhammerInf_recursion`). $\square$

Lemma 40 (Product as a series). *For $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} q^{\binom{n}{2}} z^n (q^{n+1}; q)_{\infty} = (q; q)_{\infty} (-z; q)_{\infty}.$$

Proof. Multiply the second Euler identity (Lemma 39) by the constant $(q; q)_{\infty}$ and rewrite $(q; q)_{\infty} / (q; q)_n = (q^{n+1}; q)_{\infty}$ via Lemma 40. $\square$

5.3 The key identity $S_k = 1/(q; q)_{\infty}$

This is the core of the formalisation, replacing both the coefficient-recurrence bookkeeping of Proof 1 and the complex analysis of Proof 3. After the Cauchy product is rearranged by diagonals, each diagonal coefficient is exactly one of the sums

$$S_k(q) := \sum_{m=0}^{\infty} \frac{q^{m(m+k)}}{(q; q)_m (q; q)_{m+k}},$$

and all of them collapse to $1/(q; q)_{\infty}$. This is the classical *Durfee rectangle identity*: combinatorially, $m \times (m+k)$ is the Durfee rectangle of a partition, contributing $q^{m(m+k)}$, with the regions to its right and below contributing $1/(q; q)_m$ and $1/(q; q)_{m+k}$; the $k = 0$ case is the classical Durfee square identity. (See Andrews, *The Theory of Partitions*, 1976, and Gessel, *Some generalized Durfee square identities*, Discrete Math. **50** (1984).) What is specific to this formalisation is the proof: rather than the Young-diagram bijection, the identity is established *analytically* through the contraction recurrence below, which is convenient to formalise in Lean.

Definition 41 (The key sum). For $\|q\| < 1$ and $k \in \mathbb{N}$, $S_k(q) := \sum_{m \geq 0} q^{m(m+k)} / ((q; q)_m (q; q)_{m+k})$.

Lemma 42 (Summability of S_k). *For $\|q\| < 1$, the summand of S_k is summable.*

Proof. The denominators $(q; q)_m (q; q)_{m+k}$ are bounded below in norm by a positive constant (the partial products of a convergent infinite product with nonzero limit), so the summand is dominated by $C \|q\|^{m^2}$, which is summable. $\square$

Lemma 43 (Limit of S_k). *For $\|q\| < 1$, $S_k(q) \rightarrow 1/(q; q)_\infty$ as $k \rightarrow \infty$.*

Proof. The $m = 0$ term is $1/(q; q)_k \rightarrow 1/(q; q)_\infty$ (`tendsto_qPochhammer` and non-vanishing of $(q; q)_\infty$). The tail $m \geq 1$ tends to 0 by dominated convergence: each term $q^{m(m+k)}/((q; q)_m (q; q)_{m+k}) \rightarrow 0$ as $k \rightarrow \infty$, dominated by $\|q\|^{m^2}/C$. $\square$

Lemma 44 (Recurrence for S_k). *For $\|q\| < 1$ and every k ,*

$$S_k - S_{k+1} = q^{k+1} (S_{k+2} - S_{k+1}).$$

Proof. Combine the two series term-by-term, using $(q; q)_{m+k+1} = (q; q)_{m+k} (1 - q^{m+k+1})$ to write $S_k - S_{k+1}$ as a single series, then split the numerator $1 - q^m - q^{m+k+1}$ and reindex the $(1 - q^m)$ part by $m \mapsto m + 1$. The two pieces reassemble into $q^{k+1} (S_{k+2} - S_{k+1})$. $\square$

Theorem 45 (Key identity). *For $\|q\| < 1$ and every k , $S_k(q) = 1/(q; q)_\infty$.*

Proof. Write $D_k = S_k - S_{k+1}$. Iterating Lemma 45 n times gives $\|D_k\| \leq \|q\|^{n(2k+n+1)/2} \|D_{k+n}\|$. Since $D_{k+n} \rightarrow 0$ (Lemma 44) and $\|q\|^n \rightarrow 0$, we get $D_k = 0$ for all k . Hence $S_k = S_0$ is constant in k , and its common value equals $\lim_k S_k = 1/(q; q)_\infty$. $\square$

Lemma 46 (Non-negative diagonal coefficient). *For $\|q\| < 1$ and $k \geq 0$,*

$$\sum_{m=0}^{\infty} q^{\binom{m+k}{2}} (q^{m+k+1}; q)_\infty \cdot \frac{q^{\binom{m}{2}} q^m}{(q; q)_m} = q^{\binom{k}{2}}.$$

Proof. Rewrite $(q^{m+k+1}; q)_\infty = (q; q)_\infty / (q; q)_{m+k}$ (Lemma 40) and use the arithmetic identity $\binom{m+k}{2} + \binom{m}{2} + m = \binom{k}{2} + m(m+k)$ to reduce the sum to $q^{\binom{k}{2}} (q; q)_\infty \cdot S_k(q)$. Apply Theorem 46. $\square$

Lemma 47 (Negative diagonal coefficient). *For $\|q\| < 1$ and $l \geq 0$,*

$$\sum_{n=0}^{\infty} q^{\binom{n}{2}} (q^{n+1}; q)_\infty \cdot \frac{q^{\binom{n+l+1}{2}} q^{n+l+1}}{(q; q)_{n+l+1}} = q^{\binom{l+2}{2}}.$$

Proof. As in Lemma 47, using the companion arithmetic identity $\binom{n}{2} + \binom{n+l+1}{2} + (n+l+1) = \binom{l+2}{2} + n(n+l+1)$ and the key identity $S_{l+1} = 1/(q; q)_\infty$ (Theorem 46). $\square$

5.4 Extended second Euler identity (all z)

To run the Cauchy product on the full punctured disc rather than only the annulus, the second Euler identity is extended to all $z \in \mathbb{C}$.

Lemma 48 (Series recursion). *For $\|q\| < 1$, the series $S(z) := \sum_n q^{\binom{n}{2}} z^n / (q; q)_n$ satisfies $S(z) = (1+z) S(qz)$.*

Proof. A term-by-term algebraic identity: the difference of the z - and qz -series equals $z S(qz)$, using $(q; q)_n = (q; q)_{n-1} (1 - q^n)$. Summability for all z is `qSeries.euler_second_summable_all`, proved by the ratio test (the ratio $\|q\|^n \|z\| / \|1 - q^{n+1}\| \rightarrow 0$). $\square$

Theorem 49 (Extended second Euler identity). *For $\|q\| < 1$ and every $z \in \mathbb{C}$,*

$$\sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} z^n}{(q; q)_n} = (-z; q)_{\infty}.$$

Proof. Given any z , choose N with $\|zq^N\| < 1$. Iterating Lemma 49 N times gives $S(z) = \prod_{k=0}^{N-1} (1 + zq^k) \cdot S(zq^N)$, and $S(zq^N) = (-zq^N; q)_{\infty}$ by the (small-argument) second Euler identity (Lemma 39). The product $\prod_{k=0}^{N-1} (1 + zq^k) \cdot (-zq^N; q)_{\infty} = (-z; q)_{\infty}$ by repeated use of `qSeries.qPochhammerInf_recursion`. $\square$

Lemma 50 (Extended Euler identity at q/z). *For $\|q\| < 1$ and $z \neq 0$ (no annulus condition),*

$$\sum_{m=0}^{\infty} \frac{q^{\binom{m}{2}} q^m z^{-m}}{(q; q)_m} = (-q/z; q)_{\infty}.$$

Proof. Apply Theorem 50 at the point q/z (valid for all q/z , hence all $z \neq 0$). $\square$

5.5 The triple product and bilateral series

Definition 51 (Triple product and bilateral series). $\text{jacobiProd}(q, z) := (q; q)_{\infty} (-z; q)_{\infty} (-q/z; q)_{\infty}$, and the bilateral series $\text{jacobiBilateral}(q, z) := \sum_{k \geq 0} z^k q^{\binom{k}{2}} + \sum_{m \geq 0} z^{-(m+1)} q^{\binom{m+2}{2}}$ (the non-negative and negative halves of $\sum_{k \in \mathbb{Z}} z^k q^{k(k-1)/2}$).

5.6 The Jacobi triple product

Theorem 52 (Jacobi triple product). *For $\|q\| < 1$, $\|z\| < 1$, $z \neq 0$, $\text{jacobiProd}(q, z) = \text{jacobiBilateral}(q, z)$; that is,*

$$(q; q)_{\infty} (-z; q)_{\infty} (-q/z; q)_{\infty} = \sum_{k \geq 0} z^k q^{\binom{k}{2}} + \sum_{m \geq 0} z^{-(m+1)} q^{\binom{m+2}{2}}.$$

Proof. Expand $(q; q)_{\infty} (-z; q)_{\infty}$ as a series (Lemma 41) and $(-q/z; q)_{\infty}$ via the *extended* second Euler identity (Lemma 51), which holds for every $z \neq 0$ with no lower bound on $\|z\|$. Their Cauchy product over $\mathbb{N} \times \mathbb{N}$ is summable (`Summable.mul_norm`), so by `Summable.tsum_prod` the double sum may be reordered. Partition $\mathbb{N} \times \mathbb{N}$ into the diagonals $n \geq m$ (index $k = n - m \geq 0$) and $n < m$ (index $l = m - n - 1 \geq 0$). The k -diagonal sum is $z^k q^{\binom{k}{2}}$ by Lemma 47, and the l -diagonal sum is $z^{-(l+1)} q^{\binom{l+2}{2}}$ by Lemma 48. Since the extended identity carries no annulus hypothesis, the argument runs directly on the full punctured disc $0 < \|z\| < 1$, with no separate continuation step. $\square$

Remark (divergence from the outline). Compared with Section ??: the formalisation uses a *genuine Cauchy product* of two second-Euler expansions, not Zhu’s semi-finite truncation; the diagonal coefficients are evaluated through the classical Durfee rectangle identity (Theorem 46), but via its analytic contraction-recurrence proof rather than the usual Young-diagram bijection; and the argument is carried out directly on the whole punctured disc by strengthening the Euler identity to all z (Theorem 50), so it needs neither an annulus restriction with a separate extension step, the Liouville-type complex analysis of Proof 3, nor the coefficient bookkeeping of Proof 1. (Using Durfee rectangles to reach the Jacobi triple product is itself known; see W. Chu, *Durfee rectangles and the Jacobi triple product identity*, Acta Math. Sinica **9** (1993), 24–26.)

5.7 Euler's pentagonal number theorem

Definition 53 (Generalized pentagonal number). $\omega(k) := (k(3k-1)/2)$ for $k \in \mathbb{Z}$ (the product $k(3k-1)$ is always even), realised in Lean as a natural number.

Theorem 54 (Euler's pentagonal number theorem). *For $\|q\| < 1$,*

$$(q; q)_\infty = \sum_{k \geq 0} (-1)^k q^{\omega(k)} + \sum_{k \geq 0} (-1)^{k+1} q^{\omega(-(k+1))} = 1 - q - q^2 + q^5 + q^7 - q^{12} - \dots$$

Proof. Apply the Jacobi triple product (Theorem 53) with $q \mapsto q^3$ and $z \mapsto -q$; the hypotheses reduce to $\|-q\| < 1$ and $-q \neq 0$, both immediate for $0 < \|q\| < 1$ (the case $q = 0$ is trivial). The left-hand side $(q^3; q^3)_\infty (q; q^3)_\infty (q^2; q^3)_\infty$ is regrouped into $(q; q)_\infty$ by splitting the product $\prod_{n \geq 1} (1 - q^n)$ into the three residue classes $\{3k+1\}, \{3k+2\}, \{3k+3\}$ modulo 3. Matching the right-hand exponents k^2 (with sign $(-1)^k$) against $\omega(\pm k)$ gives the pentagonal series. $\square$

5.8 Analytic extension to all $z \neq 0$

Theorem 53 was proved under the restriction $\|z\| < 1$. The module `QSeries/JTP_Analytic.lean` removes this restriction using locally uniform convergence of both sides on $\{z \neq 0\}$ and the functional equation $f(qz) = f(z)/z$ satisfied by both `jacobiProd(q, z)` and `jacobiBilateral(q, z)`.

Theorem 55 (Summability of bilateral series for all z). *For $\|q\| < 1$ and every $z \in \mathbb{C}$, the non-negative part $\sum_{k \geq 0} z^k q^{\binom{k}{2}}$ is summable.*

Proof. The ratio of consecutive terms is $|z| |q|^n \rightarrow 0$, so the ratio test gives summability; the Weierstrass M -test with $M_k = R^k \|q\|^{\binom{k}{2}}$ (summable by a further ratio test) gives uniform convergence on $|z| \leq R$. $\square$

Theorem 56 (Locally uniform convergence of the product side). *For $\|q\| < 1$, the partial products $\prod_{k < N} (1 - q^{2k})(1 + zq^{2k-1})(1 + q^{2k-1}/z)$ converge locally uniformly on $\{z \neq 0\}$ to `jacobiProd(q, z)`.*

Proof. On $\overline{B}(z_0, \|z_0\|/2)$ for any $z_0 \neq 0$: the factor $\prod (1 + (-z)q^k)$ is handled by the uniform convergence of $\prod (1 + a_k)$ on compacta (Weierstrass M -test with $\sum \|a_k\| < \infty$), and the factor $\prod (1 + (-q/z)q^k)$ uses the bound $\|-q/z\| \leq 2\|q\|/\|z_0\|$ on that ball. Products of uniformly convergent bounded sequences converge uniformly. $\square$

Theorem 57 (Locally uniform convergence of the series side). *For $\|q\| < 1$, the bilateral partial sums $\sum_{|k| < N} z^k q^{\binom{k}{2}}$ converge locally uniformly on $\{z \neq 0\}$ to `jacobiBilateral(q, z)`.*

Proof. The non-negative half is locally uniformly convergent everywhere by Theorem 56 and the M -test. The negative half $\sum_{m \geq 0} z^{-(m+1)} q^{\binom{m+2}{2}}$ is uniformly convergent on $\{\|z^{-1}\| \leq R'\}$; near any $z_0 \neq 0$ the ball $B(z_0, \|z_0\|/2)$ satisfies $\|z^{-1}\| \leq 2/\|z_0\|$. $\square$

Theorem 58 (Analytic Jacobi triple product). *For $\|q\| < 1$ and every $z \neq 0$,*

$$(q; q)_\infty (-z; q)_\infty (-q/z; q)_\infty = \sum_{k \geq 0} z^k q^{\binom{k}{2}} + \sum_{m \geq 0} z^{-(m+1)} q^{\binom{m+2}{2}}.$$

Proof. Both sides are locally uniformly convergent on $\{z \neq 0\}$ (Theorems 57 and 58). Both satisfy the functional equation $H(qz) = H(z)/z$ for all $z \neq 0$. For any fixed $z \neq 0$, choose N large enough so that $\|q^N z\| < 1$. Then Theorem 53 (valid at $q^N z$) together with the iterated functional equation for both sides yields equality at z . $\square$

6 Formal power series reformulation (algebraic proof)

This section records the second, independent proof of the Jacobi triple product that was formalised in `Qseries_Aristotle_new/QSeries/FPS.lean`, `Qseries/FPS_Euler.lean`, and `Qseries/FPS_Algebra.lean` under the namespace `qSeries.FPS`. Unlike the analytic proof of Section 5, this proof is *entirely algebraic*: no norms, no $\|q\| < 1$ hypothesis, and no topological analysis beyond the X -adic (pi) topology on $R[[X]]$.

Key idea. The variable q becomes the formal power series indeterminate $X \in R[[X]]$, while z lives in the coefficient ring $A = \text{LaurentPolynomial } \mathbb{C}$ (with $z = T(1)$, $z^{-1} = T(-1)$). The infinite q -Pochhammer symbol $(a; X)_\infty$ is a well-defined element of $R[[X]]$ because the factors $(1 - aX^k)$ converge to 1 in the pi topology: each factor affects only finitely many coefficients of any fixed degree.

6.1 FPS q -Pochhammer symbols

Throughout this section R is a commutative ring (with `DiscreteTopology`).

Definition 59 (FPS finite q -Pochhammer symbol). For $a \in R[[X]]$ and $n \in \mathbb{N}$,

$$\text{qPoch } a \ n := \prod_{k=0}^{n-1} (1 - a \cdot X^k) \in R[[X]].$$

Lemma 60 (Base case). $\text{qPoch } a \ 0 = 1$.

Proof. Empty product. □

Lemma 61 (FPS recurrence). $\text{qPoch } a \ (n + 1) = \text{qPoch } a \ n \cdot (1 - a \cdot X^n)$.

Proof. Immediate from `Finset.prod_range_succ`. □

Lemma 62 (FPS shift identity). $\text{qPoch } a \ (n + 1) = (1 - a) \cdot \text{qPoch } (aX) \ n$.

Proof. Induction on n using the recurrence. □

Lemma 63 (Coefficient stabilisation). *If $d < n$ then $[X^d] \text{qPoch } a \ (n + 1) = [X^d] \text{qPoch } a \ n$.*

Proof. The new factor $(1 - aX^n)$ only affects monomials of degree $\geq n > d$. □

Lemma 64 (Constant coefficient). *The constant term of $\text{qPoch } (X : R[[X]]) \ n$ is 1.*

Proof. Induction on n ; each factor $(1 - X \cdot X^k)$ has constant term 1. □

Theorem 65 (FPS multipliability in pi topology). *The family $k \mapsto 1 - a \cdot X^k$ is multipliable in $R[[X]]$ (with the pi / X -adic topology).*

Proof. Apply `MvPowerSeries.WithPiTopology.multipliable_one_add_of_tendsto_order_atTop_nhds_top`: the X -adic order of $-(aX^k)$ is $\geq k$ (the factor $-(aX^k)$ is divisible by X^k), so the orders tend to ∞ , which is the criterion for multipliability in the pi topology. No norm hypothesis is needed. □

Definition 66 (FPS infinite q -Pochhammer symbol).

$$\text{qPochInf } a := \prod_{k=0}^{\infty} (1 - a \cdot X^k) \in R[[X]].$$

Lemma 67 (Coefficient formula). $[X^d] \mathit{qPochInf} a = [X^d] \mathit{qPoch} a (d + 1)$.

Proof. The partial products stabilise at degree d once $n > d$ (Lemma 64), and converge to $\mathit{qPochInf} a$ in the pi topology; continuity of coefficient extraction gives the identity. $\square$

Lemma 68 (FPS recursion). $\mathit{qPochInf} a = (1 - a) \cdot \mathit{qPochInf} (a \cdot X)$.

Proof. Extract the $k = 0$ factor from the infinite product and shift: $\prod_{k \geq 0} (1 - aX^k) = (1 - a) \cdot \prod_{k \geq 0} (1 - aX^{k+1}) = (1 - a) \cdot \mathit{qPochInf} (aX)$. $\square$

Lemma 69 (Iterated telescoping). *For every $n \in \mathbb{N}$: $\mathit{qPochInf} a = \mathit{qPoch} a n \cdot \mathit{qPochInf} (a \cdot X^n)$.*

Proof. Induction on n using Lemma 69. $\square$

Lemma 70 (FPS unit). *If $1 - [X^0] a$ is a unit of R , then $\mathit{qPochInf} a$ is a unit of $R[[X]]$.*

Proof. The constant term of $\mathit{qPochInf} a$ equals $1 - [X^0] a$ (which is a unit), and a power series with unit constant term is a unit. $\square$

6.2 FPS Euler second identity

Theorem 71 (FPS Euler second identity). *In $R[[X]]$ (for any commutative ring R with discrete topology),*

$$\mathit{qPochInf}(-a) = \sum_{k=0}^{\infty} X^{\binom{k}{2}} \cdot a^k \cdot (\mathit{qPoch}(X : R[[X]]) k)^{-1}.$$

Proof. At each degree d , only finitely many terms contribute (those with $\binom{k}{2} \leq d$, i.e. $k \leq d + 1$). The key algebraic step is: $[X^j] \mathit{qBinom}(N, k, X) = [X^j] (\mathit{qPoch}(X, k))^{-1}$ whenever $j + k \leq N$ (`coeff_qBinom_eq_qPochInv`). Then the finite q -binomial theorem (Theorem 25 with $z = -a$) gives $\mathit{qPoch}(-a, N) = \sum_{k=0}^N X^{\binom{k}{2}} \mathit{qBinom}(N, k, X) (-a)^k$ in $R[[X]]$. Taking $N = d + 1$ and extracting the degree- d coefficient yields the identity at each degree. The sum is summable (each degree has finitely many nonzero contributions) and the tsum specialises to the tprod as $N \rightarrow \infty$ in the pi topology. $\square$

6.3 FPS key identity

Definition 72 (FPS key sum). For $k \in \mathbb{N}$, define

$$S_k^{\text{fps}} := \sum_{m=0}^{\infty} X^{m(m+k)} \cdot (\mathit{qPoch}(X, m))^{-1} \cdot (\mathit{qPoch}(X, m+k))^{-1} \in R[[X]].$$

Lemma 73 (FPS key sum is summable). *The summand of S_k^{fps} is summable in $R[[X]]$.*

Proof. The monomial $X^{m(m+k)}$ has X -adic order $\geq m$; since $m \rightarrow \infty$, the summand family satisfies the pi-topology criterion for summability. $\square$

Lemma 74 (FPS key sum recurrence). *For every k :*

$$S_k^{\text{fps}} - S_{k+1}^{\text{fps}} = X^{k+1} \cdot (S_{k+2}^{\text{fps}} - S_{k+1}^{\text{fps}}).$$

Proof. Combine the two series term-by-term. Write $(\mathit{qPoch}(X, m+k))^{-1} = (1 - X^{m+k+1}) \cdot (\mathit{qPoch}(X, m+k+1))^{-1}$ (`qPochInv_succ_mul`), split the numerator, and reindex the $(1 - X^m)$ -part by $m \mapsto m + 1$. This is a *purely algebraic* ring identity in $R[[X]]$; no norms are needed. $\square$

Theorem 75 (FPS: all S_k are equal). $S_k^{\text{fps}} = S_0^{\text{fps}}$ for all $k \in \mathbb{N}$.

Proof. Let $D_k = S_k^{\text{fps}} - S_{k+1}^{\text{fps}}$. By Lemma 75, $D_k = X^{k+1}D_{k+2} + X^{k+1}D_{k+1}$; iterating, $[X^d]D_k = 0$ for all $d < k+1$ (directly from the recurrence, since $[X^d]X^{k+1} = 0$ when $d < k+1$). For $d \geq k+1$, induction on d using the recurrence (strong induction) yields $[X^d]D_k = 0$ as well, because the only contributing term $[X^{k+1}]X^{k+1} = 1$ forces $[X^{d-(k+1)}]D_{k+2} = [X^{d-(k+1)}]D_k$ and the argument closes by strong induction. Hence $D_k = 0$ for all k , giving $S_k^{\text{fps}} = S_0^{\text{fps}}$. $\square$

Theorem 76 (FPS key identity). For every k ,

$$S_k^{\text{fps}} = (\text{qPochInf}(X))^{-1} =: \text{qqInv}.$$

Proof. By Theorem 76, $S_k^{\text{fps}} = S_{d+1}^{\text{fps}}$ for any d . For $k = d+1 > d$, the only term contributing to $[X^d]S_{d+1}^{\text{fps}}$ is $m = 0$ (the X^0 monomial), giving $[X^d](\text{qPoch}(X, d+1))^{-1}$. For $d+1$ steps of the recursion $\text{qPochInf} = \text{qPoch}(X, n) \cdot \text{qPochInf}(X \cdot X^n)$, one shows $[X^d](\text{qPoch}(X, k))^{-1} = [X^d]\text{qqInv}$ for $d < k$ (Lemma `coeff_qPochInv_eq_qqInv`). Hence $[X^d]S_k^{\text{fps}} = [X^d]\text{qqInv}$ for all d . $\square$

Comparison with the analytic key identity. The analytic key identity (Theorem 46) proves $S_k = 1/(q; q)_\infty$ using a *norm estimate*: $\|D_k\| \leq \|q\|^{n(2k+n+1)/2} \|D_{k+n}\| \rightarrow 0$. The FPS key identity (Theorem 77) proves the same identity *coefficient by coefficient* using only ring arithmetic in $R[[X]]$, requiring no analytic hypotheses whatsoever.

6.4 FPS Cauchy coefficient identities

Lemma 77 (FPS product expansion). In $R[[X]]$:

$$\text{qPochInf}(X) \cdot \text{qPochInf}(-a) = \sum_{n=0}^{\infty} X^{\binom{n}{2}} a^n \text{qPochInf}(X \cdot X^n).$$

Proof. Expand $\text{qPochInf}(-a)$ via Theorem 72 and distribute $\text{qPochInf}(X)$ through the tsum. The identity $\text{qPochInf}(X) \cdot (\text{qPoch}(X, n))^{-1} = \text{qPochInf}(X \cdot X^n)$ (Lemma 70) converts each term. $\square$

Theorem 78 (FPS Cauchy coefficient – non-negative diagonal). For every $k \geq 0$:

$$\sum_{m=0}^{\infty} [X^{(m+k).choose2} \cdot \text{qPochInf}(X \cdot X^{m+k})] \cdot [X^{m.choose2} \cdot X^m \cdot (\text{qPoch}(X, m))^{-1}] = X^{k.choose2}.$$

That is, the k -th diagonal sum of the Cauchy product equals $X^{\binom{k}{2}}$.

Proof. The diagonal sum equals $X^{\binom{k}{2}} \cdot \text{qPochInf}(X) \cdot S_k^{\text{fps}}$ after applying the arithmetic identity $\binom{m+k}{2} + \binom{m}{2} + m = \binom{k}{2} + m(m+k)$ (`choose2_add`) and Lemma 70. Apply Theorem 77 and $\text{qPochInf}(X) \cdot \text{qqInv} = 1$. $\square$

Theorem 79 (FPS Cauchy coefficient – negative diagonal). For every $l \geq 0$:

$$\sum_{n=0}^{\infty} [X^{n.choose2} \cdot \text{qPochInf}(X \cdot X^n)] \cdot [X^{(n+l+1).choose2} \cdot X^{n+l+1} \cdot (\text{qPoch}(X, n+l+1))^{-1}] = X^{(l+2).choose2}.$$

Proof. Analogous to Theorem 79, using the companion arithmetic identity $\binom{n}{2} + \binom{n+l+1}{2} + (n+l+1) = \binom{l+2}{2} + n(n+l+1)$ (`choose2_add'`) and $S_{l+1}^{\text{fps}} = \text{qqInv}$. $\square$

6.5 FPS Jacobi triple product

Definition 80 (FPS JTP objects). Let $A = \text{LaurentPolynomial } \mathbb{C}$, $z = T(1)$, $z^{-1} = T(-1)$ viewed as constant power series in $A[[X]]$. Define:

$$\begin{aligned} \text{jtpProd} &:= \text{qPochInf}(X) \cdot \text{qPochInf}(-z) \cdot \text{qPochInf}(-X \cdot z^{-1}) \in A[[X]], \\ \text{jtpSeries} &:= \left(\sum_{n \geq 0} T(n) \cdot X^{n.\text{choose } 2} \right) + \left(\sum_{m \geq 0} T(-(m+1)) \cdot X^{(m+2).\text{choose } 2} \right) \in A[[X]]. \end{aligned}$$

Here $T(k)$ is the degree- k monomial in the Laurent polynomial ring A , so $T(n) = z^n$ and $T(-(m+1)) = z^{-(m+1)}$.

Theorem 81 (FPS Jacobi triple product). In $A[[X]]$ (with $A = \text{LaurentPolynomial } \mathbb{C}$):

$$\text{jtpProd} = \text{jtpSeries},$$

that is,

$$\text{qPochInf}(X) \cdot \text{qPochInf}(-z) \cdot \text{qPochInf}(-X/z) = \sum_{k \geq 0} z^k X^{\binom{k}{2}} + \sum_{m \geq 0} z^{-(m+1)} X^{\binom{m+2}{2}}.$$

Proof. Expand jtpProd using Lemma 78 (on the $\text{qPochInf}(X) \cdot \text{qPochInf}(-z)$ factor) and Theorem 72 (on $\text{qPochInf}(-Xz^{-1})$), forming a double tsum over $\mathbb{N} \times \mathbb{N}$. Summability of the double product $a_n \cdot b_m$ (where $a_n = X^{\binom{n}{2}} z^n \text{qPochInf}(X \cdot X^n)$ and $b_m = X^{\binom{m}{2}+m} z^{-m} (\text{qPoch}(X, m))^{-1}$) follows from the pi-topology structure: each coefficient of $a_n \cdot b_m$ is eventually zero in n (once $\binom{n}{2} > d$) and in m (once $\binom{m}{2} + m > d$).

Partition $\mathbb{N} \times \mathbb{N}$ via the bijection $\text{diagEquiv} : (n, m) \mapsto \text{inl}(n-m, m)$ if $m \leq n$, $\text{inr}(m-n-1, n)$ otherwise. The k -diagonal ($m \leq n, k = n-m$) sum equals $z^k X^{\binom{k}{2}}$ by Theorem 79; the l -diagonal ($m > n, l = m-n-1$) sum equals $z^{-(l+1)} X^{\binom{l+2}{2}}$ by Theorem 80. Summing over all k and l gives jtpSeries . $\square$

Comparison with Section 5.

- The FPS proof (Theorem 82) works over *any* commutative ring with discrete topology and needs no $\|q\| < 1$ hypothesis; the analytic proof (Theorem 53) works in $\mathbb{C}$ with $\|q\| < 1$ and $\|z\| < 1$.
- The analytic proof is extended to all $z \neq 0$ by Theorem 59; the FPS proof is already valid for all $z \in A^\times$.
- Both use the same algebraic infrastructure (Sections 4–4.3). The key identity ($S_k = (q; q)_\infty^{-1}$) is proved analytically in Theorem 46 and algebraically (in $R[[X]]$) in Theorem 77.